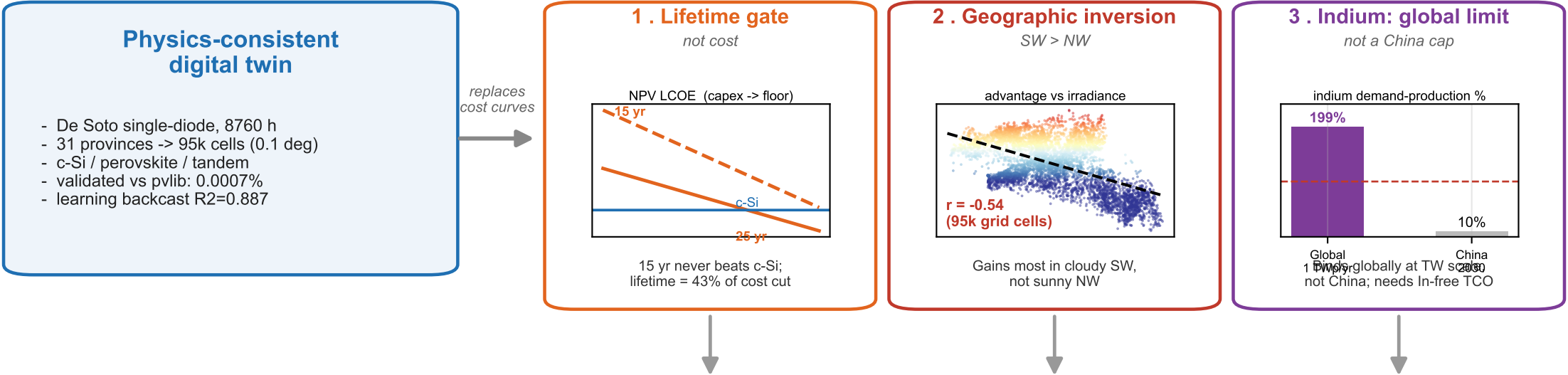
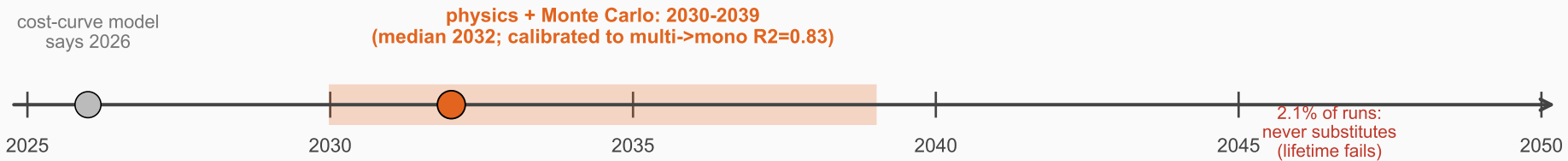


Physics rewrites the perovskite substitution narrative

An 8760-h physics-consistent digital twin of China's PV transition reveals three structural shifts cost curves miss



Synthesis: when does substitution actually happen?



Policy: shift R&D from cost to lifetime . site perovskite in the Southwest, not the Northwest . pursue indium-free electrodes for the TW endgame